(ODE variable coefficients)

Power series — Frobenius meth

$P_0\frac{d^2y}{dx^2} + P_1\frac{dy}{dx} + P_2 y = 0$

$\frac{d^2y}{dx^2} + \frac{P_1}{P_0}\frac{dy}{dx} + \frac{P_2}{P_0}y = \frac{R}{P_0}$

$\frac{d^2y}{dx^2} + \frac{dy}{dx} + xy = 0$

$y = \sum a_n x^n$

$\frac{dy}{dx} = \sum a_n n x^{n-1}$

$\frac{d^2y}{dx^2} = \sum a_n n(n-1)x^{n-2}$

$\sum_{n=1} a_n n x^{n-1} + x\sum a_n x^n + \sum_{n=1} a_n x^n = 0$

$a_1 x^0 + a_2 2x^1 + a_3 3x^2 + \dots + a_n n x^{n-1} + a_0 x^1 + a_1 x^2 + a_2 x^3 + \dots + a_{n-1}x^n + \dots$

$x^0(x) + x^1(2a_2 + a_0) + x^2(3a_3 + a_1) + x^3(4a_4 + a_2) + \dots$

x^0: $a_1 = 0$

x^1: $a_2 = -a_0$

x^2: $a_3 = -\frac{a_1}{3} = 0$

x^3: $a_4 = \frac{a_2}{4}$

x^4: $a_5 = \frac{a_3}{5} = 0$

$y = \sum a_n x^n$

$= a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \dots$

$= a_0\left(1 - \frac{x^2}{2} + \frac{x^4}{8} + \dots\right)$

$= a_0 y_1 + a_1 y_2$

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regular / Analytic fn

if developed as power series $\frac{1}{1-x} = \sum x^n$, $|x| < 1$

$\frac{2x}{1-x} = 2x\sum x^n$, $\frac{2x}{1-x^2} = 2x\sum x^{2n}$ } regular

$\frac{d}{dx}$, $\int$ } product $\frac{1}{1-x}$, $\frac{x}{1-x}$ Stark $|x| < 1$

Ordinary pt / regular pt DE

$\frac{d^2y}{dx^2} + P_1(x)\frac{dy}{dx} + P_2(x)y = 0$

$x=0$ ordinary pt if P_1, P_2 both regular fn

$\Rightarrow$ sol is again $y = \sum a_n x^n$

$\frac{d^2y}{dx^2} - xy = 0$

$P_1(x) = -x$ is regular can be in power

$\therefore x=0$ ordinary pt

$y = \sum a_n x^n$

$y' = \sum a_n n x^{n-1}$

$y'' = \sum a_n n(n-1)x^{n-2}$

$\sum_{n=2} a_n n(n-1)x^{n-2} - x\sum_{n=0} a_n x^{n+1} = 0$

x^0: $a_2 2 x^0 + a_3 6x^1 + a_4 12x^2 + a_5 20x^3$

$-[a_0 x^1 + a_1 x^2 + a_2 x^3 + \dots] = 0$

$2a_2 x^0 + x^1[6a_3 - a_0] + x^2[12a_4 - a_1] + x^3[20a_5 - a_2] + \dots$

x^0: $a_2 = 0$

$a_0 = 6a_3$

$a_1 = 12a_4$

$a_2 = 20a_5 = 0$

$y = \sum a_n x^n$

$= a_0 + a_1 x + a_2 x^2 + \dots$

Singular pts

$P_0(x) = 0 \Rightarrow x = 0$ singular pt

$P_1\frac{d^2y}{dx^2} + P_1\frac{dy}{dx} + P_2 y = 0$

$\frac{d^2y}{dx^2} + P_1\frac{dy}{dx} + P_2 y = 0$

$P_1, P_2 \to \infty$

✓ remain

✗ rm

if $P_1, P_2 \to c\infty$ as $x \to 0$ — regular singular pt — at least power series sol'n

irregular singular pt — X DNE

Summary: $P_0\frac{d^2y}{dx^2} + P_1\frac{dy}{dx} + P_2 y = 0$

if $x=0 \wedge P_0 \neq 0 \Rightarrow x=0$ ordinary pt $\Rightarrow y = \sum a_n x^n$

if $x=0 \wedge P_0 = 0 \Rightarrow x=0$ singular pt

$x \to 0$, $P_0 \to 0$

check $P_1 = \frac{P_1}{x}x$, $P_2 = \frac{P_2}{x^2}x^2$ → finite

check x: $P_1 = \frac{P_1}{x}x$, $P_2 = \frac{P_2}{x^2}x^2$ → finite

if you can: $x=0$ regular singular pt $\Rightarrow y = \sum a_n x^{mc}$

else irregular $\Rightarrow$ DNE

check ⊙ → regular / not

P_1, P_2 both has to be regular

$2x(1-x)\frac{d^2y}{dx^2} + (5-7x)\frac{dy}{dx} - 2y = 0$

$P_0 = 2x(1-x) \to 0$ as $x \to 0$

$\therefore x=0$ is a singular pt

check: $\frac{d^2y}{dx^2} + \underbrace{\frac{5-7x}{2x(1-x)}}_{P_1}\frac{dy}{dx} - \underbrace{\frac{2}{2x(1-x)}}_{P_2}y = 0$

$\lim_{x\to 0} x \cdot P_1 = \lim_{x\to 0} x\frac{5-7x}{2(1-x)} \to$ finite

$x^2 \cdot P_2 \to$ finite

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frobenius fn

y point $x=0$ singular $\to$ regular

y $x=0$ for c? ← lowest $\lim_{x\to 0} x^c$ equal

y $x=0$?

$2x^2\frac{d^2y}{dx^2} - x\frac{dy}{dx} + (x^2+1)y = 0$

$P_0(x) = 2x^2 \to 0$

$\therefore x=0$ is a singular pt

$\frac{d^2y}{dx^2} - \underbrace{\frac{1}{2x}}_{P_1}\frac{dy}{dx} + \underbrace{\frac{(x^2+1)}{2x^2}}_{P_2}y = 0$

check: $\lim_{x\to 0} x \times P_1 = \lim_{x\to 0} \frac{x \cdot 1}{2x} \to -\frac{1}{2} < \infty$

$\lim_{x\to 0} x^2 \times P_2 = \lim_{x\to 0} x^2\left(\frac{x^2+1}{2x^2}\right) = \frac{1}{2} < \infty$

$\Rightarrow x=0$ is a regular singular point

$\Rightarrow$ sol'n $y = \sum a_n x^{mc}$

$y' = \sum a_n(mc)x^{(mc-1)}$

$y'' = \sum a_n(mc-1)(mc)x^{(mc-2)}$

sub back:

$2x^2\sum a_n(mc-1)(mc)x^{(mc-2)}$

$-x\sum a_n(mc)x^{(mc-1)}$

$+(x^2+1)\sum a_n x^{mc} = 0$

$\sum a_n(mc-1)(mc)x^{mc}$

$-\sum a_n(mc)x^{mc}$

$+\sum a_n x^{mc+2} + \sum a_n x^{mc} = 0$

$\sum x^{mc}[2a_n(mc-1)(mc) - a_n(mc) + a_n]$

$+\sum a_n x^{mc+2} = 0$

$n=2=k$

$k=2$

$\frac{n=0}{x^c}$: $2a_0(c-1)c - a_0 c + a_0 = 0$

$a_0[2c(c-1) - c + 1] = 0$

$a_0 \neq 0$ const

$2c^2 - 2c - c + 1 = 0$

$2c^2 - 3c + 1 = 0$

$(2c-1)(c-1) = 0$

$c = \frac{1}{2}, 1$

$2c = 1$

$c = 1$

$\frac{n=1}{x^{c+1}}$: $2a_1(c)(c+1) - a_1(c+1) + a_1 = 0$

$a_1[2c^2 + c - 1] = 0$

$\neq 0$

$\frac{x^{n+c+2}}{}$: $\sum x^{n+c+2}\left[a_n + 2a_{n+2}(n+c-1)(n+2+c)\right] = 0$

$a_n + a_{n+2}\left[-(n+c+2)+1\right] = 0$

$a_{n+2} = \frac{-a_n}{2(n+c-1)(n+c+2) - (n+c+2) + 1}$

$2[n^2 + cn + 2n + cn - c^2 - 2c - n - c - 2] - n - c - 1$

$n=0$: $a_2 = \frac{-a_0}{-c^2 - 4c - 5}$

$2n^2 + 4cn + 2n - c^2 - 4c - 5$

$n=1$: $a_3 = \frac{-a_1}{\text{--}} = 0 = a_5 = a_7$

$n=2$: $a_4 = \frac{-a_2}{4 + 8c + 2 - c^2 - 4c - 5}$

$-c^2 + 4c + 1$

$\vdots$

$y = \sum a_n x^{n+c}$

$= a_0 x^c - \frac{a_0}{-c^2 - 4c - 5} + \frac{a_0}{(c^2 - 4c - 5)(-c^2 + 4c + 1)} + \dots$

$c = 1$

Ordinary points

first take every thing to y'' x^n

$\sum_{n=2} a_n n x^{n-2} y$

$\sum_{n=2-2} a_{n+2}(n+2)x^n$

$\Downarrow$

then take n equal

frobenius

$\sum_{n=0} y\, y'\, y''$ for all

k meth

or